CompliQ Project Feature List

CompliQ is a web application for managing renewable-energy inspection data, currently organized around wind-turbine and solar-array workflows. This list describes capabilities found in the project source; it distinguishes database-backed functions from demo or browser-local behavior where that matters.

User-facing features

Public website

- Product landing page with platform overview, feature highlights, navigation, theme controls, and demo/contact form.
- Entry points to sign in, create an account, and launch the platform.
- Wind and solar inspection product positioning.

Authentication and organizations

- Sign in and account registration with email, password, organization, and sector selection.
- Organization selection populated from the backend organization list.
- Wind or solar dashboard selection during authentication.
- Password visibility controls, password-strength feedback, and a forgot-password interface.
- Newly registered accounts are marked pending until an administrator approves them.
- Server-side password hashing, expiring bearer-token sessions, logout, and account status checks.
- Global administrator sign-in is selected through the Admin organization option.

Wind fleet dashboard

- Fleet overview metrics and health summaries.
- Turbine registry with filters, search, asset status, and detail views.
- Inspection and defect records with severity, evidence, and maintenance/dispatch decisions.
- Interactive turbine/blade inspection visualization with defect hotspots and RGB/thermal imagery options.
- Charts for fleet health, defect categories, severity, and inspection history.
- Manual turbine and defect entry, including CSV import assistance.
- Telemetry/MQTT-style status and event-log interface.
- Natural-language query/copilot interface for asking about fleet data.
- PDF inspection and engineering reports.
- Organization-specific fleet data storage in MongoDB.

Solar fleet dashboard

- Solar-array fleet overview with health, status, and telemetry panels.
- Array registry, filters, detail view, and array thermal schematic.
- Defect records with severity, engineering assessment, and maintenance/dispatch decisions.
- Charts for fleet health, defect categories, severity, and DC power output.
- Manual solar-array and defect entry, file upload, and records log.
- Natural-language query/copilot interface.
- Downloadable PDF array and inspection reports.
- Organization-specific solar dashboard access and solar-specific admin tools.

Account settings

- Profile details and avatar management.
- Organization/role information and document management.
- Password change, security preferences, notifications, and display preferences.
- Active-session view, data export, and account deletion controls.

Administrator features

Global administration

- Platform overview counters for users, uploads, reports, organizations, recently active users, pending approvals, and reports per user.
- User and organization summaries, analytics charts, signup/activity views, and CSV export.
- Account creation, role changes, account removal, and pending registration approval or rejection.
- Organization creation with a default Fleet Operator account and a clean, empty fleet-data structure.
- Organization removal, including associated users, sessions, and organization data.
- Access-control and activity/audit views.
- Dashboard metrics loaded from the protected admin API and refreshed while the page is open.

Solar administration

- Solar user and organization administration.
- Solar-focused overview, role/access information, activity/audit views, and approval controls.

Backend and data management

- Express API backed by MongoDB Atlas.
- Collections for users, organizations, sessions, fleet data, and activity events.
- Organization and sector scoping for user identity and fleet-data records.
- Authenticated APIs for account profile/password changes and account deletion.
- Authenticated APIs for reading and saving approved fleet-data keys.
- Admin-protected APIs for account management, organization management, metrics, and approvals.
- Activity events for uploads and report generation.
- Static hosting of the public application from the public directory.

Current implementation limits to keep in mind

- The wind and solar dashboards include demo/sample inspection and telemetry content; the presence of a live-looking panel does not itself mean it is connected to a real turbine, SCADA, or MQTT broker.
- The AI/copilot interfaces are implemented in frontend code and should be treated as local/demo query behavior unless a separate AI service is configured.
- File import and image interactions are frontend workflows; the repository does not show a full cloud object-storage pipeline for uploaded inspection media.
- Upload/report activity counts are recorded going forward by activity-event calls. Historical files/reports held only in browser storage are not automatically imported into MongoDB.
- Some profile preferences, avatar/document state, and dashboard interactions may remain browser-local rather than shared server-side data.
- The project provides account and organization deletion endpoints; production use should pair destructive admin actions with backups, audit retention, and recovery policies.

Main source areas

- index.html, landing.js, landing.css — public landing page.
- auth.html — sign-in, registration, organization/sector selection.
- dashboard.html, dashboard.js, dashboard.css — wind fleet dashboard.
- solar-dashboard.html — solar fleet dashboard.
- account.html — account settings.
- admin.html, admin-solar.html — global and solar admin dashboards.
- backend/server.js, backend/database.js — API, authentication, and MongoDB persistence.